

SET-A

Patan Multiple Campus
Bachelor in Information Technology (BIT)
Pre-Board Exam-2083

Time: 2 hours (10:00 AM to 12:00 PM)
Bachelor Level/3rd Year/Sixth Semester/Science
Subject: NET Centric Computing (BIT351)

Total Marks: 60
Pass Marks: 24

Group-A

Attempt any TWO questions. $[2 \times 10 = 10]$

1. Explain the complete execution process of a .NET application. Discuss CLI, MSIL, CLR, JIT compiler, and conversion into native code with a suitable diagram.
2. Explain ADO.NET architecture in detail. Describe Connection, Command, DataReader, and DataAdapter objects. Write an ADO.NET program to retrieve records from a BOOK table where price is greater than 500.
3. Explain URL routing in ASP.NET Core. Compare conventional routing and attribute routing. Explain how an HTTP request is mapped to a controller action.

Group-B

Attempt any Eight questions. $[8 \times 5 = 40]$

4. Differentiate between abstract class, interface, and sealed class in C#.
5. Explain asynchronous programming using async and await keywords.
6. Explain delegates and events in C# with examples.
7. Explain data models and DbContext in Entity Framework Core.
8. What are Action Result types in ASP.NET Core MVC? Explain with examples.
9. Explain JSON and its role in ASP.NET Core Web API.
10. Explain user-defined exceptions in C#.
11. Explain caching as a server-side state management technique.
12. Explain the procedure for publishing an ASP.NET Core application to Microsoft Azure.

Tribhuvan University
Faculty of Science and Technology
Preboard Exam
2026

SET -Q

Subject: Database Administration (BIT 352)

F.M.: 60

Group-A

Long Answer Questions

Answer any TWO Question: [2*10 =20]

1. Explain about resource management and resource manager in Oracle. Discuss about scheduler components in detail. [4 + 6]
2. Explain the storage structure in Oracle. List different administrative tools available to DBA with explanations. [5+4]
3. Discuss the roles and responsibilities of a DBA? Explain about tablespace and datafile in oracle. How we can add new tablespace and datafiles in oracle database? Explain with suitable commands. [4+2+4]

Group-B

Short Answer Questions

Answer any EIGHT Question. [8*5 =40]

4. How can you use Data Pump to import and export data? Explain [2 +3].
5. How ARCHIVELOG and NOARCHIVELOG modes are different? Explain the use of data recovery advisor. [2 +3]
6. Explain about Enterprise Manager Support Workbench. [5]
7. What is privilege? Write the commands to create user, grant privilege, revoke privilege and drop user in oracle database. [1 + 4]
8. How resource management is done using profile? List any three resource parameters with their significances. [2+3]
9. Explain oracle multitenant architecture with reference to CDB and PDB.[5]
10. What is audit trail? Explain about AUDIT_ADMIN and AUDIT_VIEWER Roles. [2+3]
11. What is performance management? Explain about performance planning in database. [2+3]
12. Write short notes on:
 - a) Row chaining and Migration
 - b) AWR in Oracle

Patan Multiple Campus
Bachelor in Information Technology
Pre-Board Examination-2083

Subject: Management Information System
Time: 3 hrs.

F.M.: 60
P.M.: 24

SET-B
Group 'A'

Attempt any Two Questions

[10*2=20]

1. What are key challenges of Management Information System?
✓ Explain merits and demerits of globalization.
2. Explain management information system. Differentiate between
✓ Decision support system (DSS) and Executive information system.
3. What is Technical definition of organization? Differentiate between
organization
& Information System.

Group 'B'

Attempt any Eight questions

[8*5=40]

- ✓ 4. How IS impacts organizations and Business Firms? Explain
- ✓ 5. Explain supply chain management system with example.
- ✓ 6. What is collaboration? How business firm can be collaborated?
- ✓ 7. Differentiate between data mart and data warehouse.
8. Explain business value chain model.
- ✓ 9. Explain moral dimensions of MIS.
- ✓ 10. What is organizations? Explain features of organizations.
- ✓ 11. Illustrate business intelligence? Explain its features.
12. Define business process reengineering. Explain.
- ✓ 13. Explain knowledge based system with example.

PATAN MULTIPLE CAMPUS

Pre-board Exam - 2083

Bachelor Level/Third Year/Six Semester
Bachelor in Information Technology (RSM 354)
(Research Methodology)

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Set-A

Section – A

Attempt any TWO questions.

(2 × 10 = 20)

1. Define and discuss case study method. Describe the different steps of case study approach and explain the limitation of case study.
2. Define research objectives. Discuss its types. Discuss how the research objectives are related to the research questions?
3. Define population, sample, sampling and sampling frame. Discuss the sampling error and sampling error. Define questionnaire. What are the types of questionnaire and describe its types.

Section – B

Attempt any EIGHT questions.

(8 × 5 = 40)

4. What do you mean by concept of research? What is the advantage of concept of research?
5. Define exploratory research design and discuss the major types of qualitative techniques for exploratory research design.
6. Define research hypothesis. Describe what are null and alternative hypothesis?
7. Why needs of literature review. What are outline of literature review? Discuss the importance of literature reviews.
8. Write the steps involved in research process?
9. What are reliability and validity? Define physical scale. Describe different types of physical scale.
10. Write advantage and disadvantage of descriptive research.
11. Determine the minimum sample size required so that the sample estimate lies within 10% of the true value with 95% level of confidence when coefficient of variation is 60%.
12. Write short notes
 - Discuss the methods of simple random sampling.
 - Discuss the methods of collecting primary data.

BEST OF LUCK!!!

Patan Multiple Campus
Bachelor in Information Technology
Pre-Board Examination-2083

Subject: Multimedia Computing
Time: 3 hrs.
6th Semester

F.M.: 60
P.M.: 24

SET A

Long Questions:

Attempt All

[10*2=20]

1. What is computer-based animation? Explain animation languages, timeline-based and frame-based animation, tween-based animation, methods of controlling animation, display of animation, and transmission of animation.
2. What is data compression? Explain the basic principles of compression and discuss the importance of compression in multimedia systems with suitable examples.

Short Questions:

Attempt any Eight

[5*8=40]

3. Explain digital video and video signal representation in detail.
4. Calculate the file size in bytes for a 60-second stereo audio recording sampled at 42.1 kHz with 16-bit resolution.
5. Discuss the factors that should be considered when selecting a compression technique for multimedia data.
6. Explain the major phases of multimedia development.
7. Explain video signal representation in detail. Compare analog and digital representations.
8. Explain the major principles of designing an effective multimedia user interface.
9. Consider the message:
AABBBCCDDDDDD
The frequencies are:

Symbol	Frequency
A	2
B	2
C	3
D	5

Construct the Huffman tree, Generate the Huffman code for each symbol and calculate total bits before coding, after coding and percentage of space save.

END